Superpixel Based Speckle Noise Filtering for Video SAR-GMTI

BoWen Jiang
Hangzhou Institute of Technology
Xidian University
Hangzhou, China
24241215109@stu.xidian.edu.cn

Dengshuai Zhao
State-Owned Luoyang Dancheng Radio Factory
Luoyang, China
21021211020@stu.xidian.edu.cn

Min Tian
Hangzhou Institute of Technology
Xidian University
Hangzhou, China
tianmin@xidian.edu.cn

PengHui Huang
Shanghai Jiao Tong University
Shanghai, China
huangpenghui@sjtu.edu.cn

Zhiwei Yang
National Key Laboratory of Radar Signal Processing, Xidian University
Xi'an, China
yangzw@xidian.edu.cn

Guisheng Liao
Hangzhou Institute of Technology
Xidian University
Hangzhou, China
liaogs@xidian.edu.cn

***Abstract*—Video synthetic aperture radar (SAR) systems present substantial prospects in ground moving target indication (GMTI) due to high frame rate and fine resolution of SAR images. In this paper, a novel speckle noise filtering method that exploits the magnitude and spatial statistics of superpixels across SAR image frames is proposed. The method aims to suppress speckle noise and enhance the gray contrast of moving objects—comprising moving targets and their associated shadows—against the background. In this method, the simple linear iterative clustering (SLIC) algorithm is first employed to segment the SAR image into superpixels based on the pixel intensity and spatial similarities in each frame. Subsequently, a centroid-based inter-frame registration scheme is developed to align the spatial position of dynamic superpixels across frames. Based on the results, a speckle noise filter is formulated with the independent and identically distributed (i.i.d.) samples determined from superpixels in adjacent frames. Experiments on real-world video SAR datasets provided by Sandia National Laboratories demonstrate that the proposed method can effectively suppress speckle noise and preserve the details of moving objects, increasing the signal-to-background contrast.**

***Index Terms*—Video SAR, speckle filtering, superpixel, dynamic registration**

I. Introduction

Video Synthetic Aperture Radar (SAR) [1]-[3] can acquire a high-frame-rate sequence of high-resolution SAR images, irrespective of weather conditions or time of day, which has shown an increasing application in wide-area surveillance [4]. SAR technology is a coherent computational imaging process, speckle noise inherently occurs due to system nonlinear errors, multi-path scattering and some other factors [5], which leads to an increased probability of false alarms in ground moving target indication (GMTI) and reduces the reliability for subsequent surveillance and tracking. Therefore, effective speckle noise filtering is necessary to address this problem.

Current research on video SAR speckle noise filtering can be broadly categorized into two main approaches. The first approach involves filtering each frame separately based on single-image information [6]. Within the spatial domain, traditional filters such as the Lee, Frost and Enhanced Lee filters [7] reduce speckle noise by performing statistical smoothing within local neighborhood windows. Furthermore, the Non-Local Means (NLM) filter [8] exploits non-local self-similarity to increase the number of effective samples, thereby significantly improving the filtering results. However, these single-frame methods are often constrained by limited sample numbers, leading to insufficient speckle noise suppression or the loss of subtle textures. The second approach adapts optical video filtering techniques such as the Video-Block Matching and 3D filtering (V-BM3D) algorithm [9], which performs collaborative filtering in a 3D transform domain. Recently, 3-Dimensional Context Covariance Matrix (3D-CCM) filter [10] was proposed, which utilizes sequential video SAR images to obtain more independent and identically distributed (i.i.d.) training samples and constructs covariance matrices to characterize their scattering statistics. By performing similarity tests on these matrices to select similar samples and subsequently applying a mean filter for estimation, the method achieves robust speckle suppression and effectively enhances the filtering performance in stationary backgrounds. However, 3D-CCM relies on a stationary scatterer assumption, causing performance to deteriorate in dynamic scenarios. Without accurate prior knowledge of target trajectories, motion-induced misalignment among frames cannot be effectively compensated in 3D-CCM. Consequently, covariance-based averaging mixes moving targets and their shadows with background clutter, resulting in severe motion blur and shadow edge degradation.

To address these issues, this article proposes an enhanced speckle noise filter based on superpixel dynamic registration. Firstly, the superpixel images are constructed using the Simple Linear Iterative Clustering (SLIC) [11]-[13] algorithm according to pixel intensity and spatial distribution characteristics. Subsequently, a centroid-based inter-frame registration scheme is proposed to align the spatial positions of dynamic superpixels across frames.Based on the alignment, a speckle noise filter is formulated with the i.i.d. samples determined from superpixels in adjacent frames. Finally, experiments are conducted using the video SAR datasets provided by Sandia National Laboratories and the results validate the improved performance of the proposed method.

Notations: $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$ are the operators of transpose and conjugate transpose, respectively. For a complex scalar a, $|a|$ represents its modulus. For a square matrix $\mathbf{A}$, $|\mathbf{A}|$ denotes its determinant. Additionally, $\|\mathbf{g}\|_2$ represents the Euclidean norm of a vector $\mathbf{g}$.

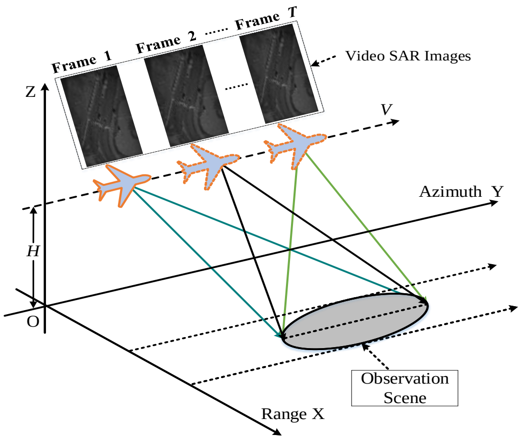

Fig. 1. Space geometry model of observation in airborne video SAR spotlight imaging mode.

II. Signal Model

Consider an airborne video SAR system operating in the spotlight mode, where the airborne platform moves along the azimuth direction (Y-axis) at a constant velocity V and height H ,while the line-of-sight projection corresponds to the range direction (X-axis). The observation geometry is illustrated in Fig. 1. Suppose the system transmits Linear Frequency Modulation (LFM) pulses and receives the echoes. After performing SAR imaging, platform motion compensation, image registration and calibration, T SAR images with well-aligned coordinates are obtained for each SAR. Next, these SAR images are matched spatially based on their positions. For the t-th frame, the complex signal in the pixel k is denoted as $z_t(k)$, where $k=1,2,\cdots,K$, $t=1,2,\cdots,T$. To facilitate the identification of moving objects (comprising both the moving targets and their associated shadows), a ternary hypothesis test for pixel k is defined as follows:

$$\begin{cases} H_0: z_t(k)=c_t(k)m_t(k)+e_t(k), \\ H_1: z_t(k)=[s_{t,tar}(k)+c_t(k)]m_t(k)+e_t(k), \\ H_2: z_t(k)=[s_{t,sha}(k)+c_t(k)]m_t(k)+e_t(k), \end{cases} \quad (1)$$

where H_0 is the target-absent hypothesis, H_1 is the target-present hypothesis and H_2 is the target-shadow hypothesis, respectively. The complex scattering components of the moving target signal $s_{t,tar}(k)$, its associated shadow signal $s_{t,sha}(k)$ and clutter signal $c_t(k)$ are modeled as zero-mean Circular Complex Gaussian (CCG) processes; $e_t(k)$ represents the additive thermal noise following a Gaussian distribution and speckle noise $m_t(k)$ is modeled as a multiplicative random variable following a Gamma distribution with unit mean, reflecting the coherent nature of SAR imaging.

III. Proposed Filter

In this section, we present a superpixel based speckle noise filtering for video SAR-GMTI. Fig. 2 illustrates the flowchart of the proposed method. In brief, the input video synthetic aperture radar image sequence is first subjected to image registration. Subsequently, the SLIC algorithm is utilized to partition the images into superpixels that encapsulate localized scattering features. To address inter-frame misalignment of moving objects, dynamic inter-frame registration is implemented at the superpixel level by aligning their spatial positions across the sequence. Finally, a covariance matrix is constructed to characterise local scattering statistics. By averaging each image pixel using similar i.i.d. samples, the filtering estimation values are obtained, achieving robust spatiotemporal speckle noise filtering. The proposed method is described in detail in the following section.

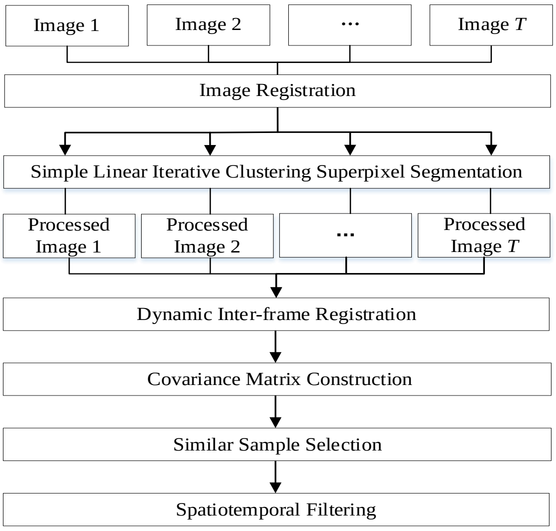

Fig. 2. Flowchart of the proposed superpixel-based speckle noise filtering method.

A. Superpixel Segmentation

SLIC performs superpixel segmentation on the video SAR image sequence by clustering pixels using a distance measure that combines intensity similarity and spatial proximity. The K pixels in each video SAR frame are aggregated into N superpixel blocks. Each superpixel block is associated with a cluster center n. For a pixel k and a cluster center n within the same frame t, the distance metric is formulated as:

$$D_t(k,n)=\sqrt{d_{\mathrm{I},t}^{\,2}(k,n)+\left(\frac{d_{\mathrm{s},t}(k,n)}{S}\right)^2\gamma^2}, \quad (2)$$

where $d_{\mathrm{I},t}(k,n)=|I(k)-I(n)|$ and $d_{\mathrm{s},t}(k,n)=\|\mathbf{g}(k)-\mathbf{g}(n)\|_2$ represent the intensity distance and spatial Euclidean distance, respectively. $I(k)$ and $I(n)$ denote the intensity of the pixel and the cluster center; $\mathbf{g}(\cdot)=[x,y]^{\mathrm{T}}$ represents the coordinate vector, x and y denote the range and azimuth coordinates; $S=\sqrt{K/N}$ represents the average distance between adjacent centers and γ (typically within [1, 40]) is the compactness parameter used to weight the intensity distance and spatial distance. Based on this metric, the cluster centers are iteratively updated until convergence, partitioning the image into disjoint superpixel regions.

B. Dynamic Inter-frame Registration

Following superpixel segmentation, the generated superpixel blocks can be categorized into two types of regions: dynamic regions and the stationary background. As illustrated in Fig. 3, consider the b-th superpixel block $B_t(b)$ (red box) in frame t as a representative dynamic region. For such dynamic

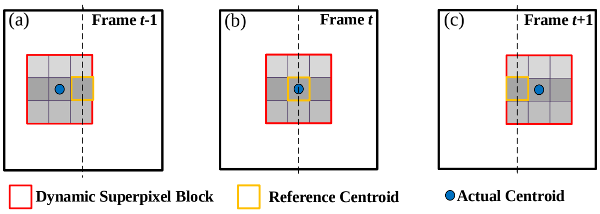

Fig. 3. Schematic diagram of the dynamic inter-frame registration principle, where (a) is for frame t-1 , (b) is for frame t , (c) is for the frame t+1 .

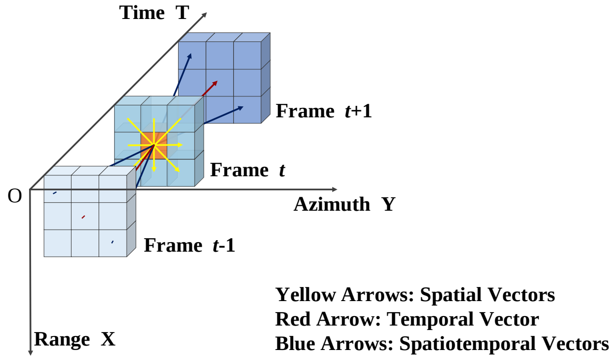

Fig. 4. Schematic diagram of the 3 × 3 × 3 spatiotemporal neighborhood structure.

regions, the superpixel position varies across frames. The actual centroid of $B_t(b)$ in the reference frame t is taken as the reference centroid (yellow box). To associate independent superpixels across frames, the coordinate of this Reference Centroid is used as a spatial probe; the superpixel in the adjacent frame that encompasses this coordinate is identified as the corresponding match. The geometric center of this matched block is defined as the actual centroid (blue dot). Due to target motion, a positional deviation arises between these two centroids in adjacent frames. The registration is achieved by compensating for the spatial offset vector calculated between the actual and reference centroids, thereby aligning the dynamic superpixel blocks to the reference position. In contrast, for stationary background superpixels, the centroid coordinates remain consistent, and no correction is required. Consequently, all superpixel blocks are spatially aligned in the superpixel domain.

C. Superpixel Based Speckle Noise Filtering

Speckle noise filtering is performed on sequence images aligned in the superpixel domain. To effectively suppress speckle noise while preserving fine texture, the filtering process evaluates each pixel to be filtered by utilising both local and global contextual information. For the pixel k to be filtered in frame t, in order to accurately characterise the scattering behavior of this pixel, a local 3 × 3 × 3 (range × azimuth × time) neighborhood window is used to construct the direction vectors $\mathbf{v}_{t,l}(k)$. As shown in Fig. 4, thirteen directional vectors are constructed by sampling complex-valued pixel intensities along spatial, temporal and spatio-temporal paths that pass through the center pixel k.

More generally, for a local $W \times W \times \delta$ neighborhood window, a set of L directional vectors are constructed. To accurately characterize local spatiotemporal scattering statistics while ensuring the estimated covariance matrix is full-rank and statistically robust, the sequential image covariance matrix of the pixel k is formulated as:

$$\tilde{\mathbf{R}}_t(k) = \frac{1}{W^2}\sum_{w\in\Omega}\mathbf{R}_t(w), \tag{3}$$

where $\mathbf{R}_t(w) = \frac{1}{L}\sum_{l=1}^{L}\mathbf{v}_{t,l}(w)\mathbf{v}_{t,l}^{\mathrm{H}}(w)$, Ω denotes the set of pixels within the neighborhood of pixel k (including k itself), W^2 is the total number of pixels in Ω .

Subsequently, to identify similar samples, the logarithmic form of the likelihood ratio $Q_t(k,k_c)$ is employed to evaluate the similarity between the pixel k and any pixel k_c within the aligned spatiotemporal superpixel block that contains k :

$$\begin{aligned}\ln Q_t(k,k_c) = 2\delta\ln 2 + \ln\left|\tilde{\mathbf{R}}_t(k)\right| + \ln\left|\tilde{\mathbf{R}}(k_c)\right| \\ -2\ln\left|\tilde{\mathbf{R}}_t(k) + \tilde{\mathbf{R}}(k_c)\right|,\end{aligned} \tag{4}$$

where δ represents the dimension of the $\mathbf{R}_t(w)$, $\tilde{\mathbf{R}}_t(k)$ and $\tilde{\mathbf{R}}(k_c)$ represent the covariance matrices of the pixel k and the pixel k_c, respectively. For the pixel k_c, if the condition $\ln Q(k,k_c) \ge T_{\mathrm{th}}$ is satisfied, it is identified as a similar i.i.d. sample, where T_{th} is a predefined similarity threshold. After evaluating all pixels within the block across the registered sequence, assume that a total of $U(k)$ samples fulfill this criterion, these identified samples (including the pixel k) are aggregated into a corresponding spatiotemporal set $\Phi(k)$. Finally, the filtered estimate $\hat{p}_t(k)$ of the pixel k in frame t is calculated by averaging the magnitudes of all samples to enhance shadow contrast, as the shadow-based GMTI primarily exploits intensity-domain signatures:

$$\hat{p}_t(k) = \frac{1}{U(k)}\sum_{u\in\Phi(k)}\left|p(u)\right|, \tag{5}$$

where $p(u)$ represents the complex pixel value of the sample u .

IV. Experimental Results and Analysis

To evaluate the proposed method, experiments were conducted using the real video SAR datasets released by Sandia National Laboratories, focusing on Frames 211 to 213. In video SAR imagery, shadows of moving targets are prioritized for performance evaluation because they retain high geometric fidelity, whereas the targets themselves often appear as defocused signatures due to Doppler modulation. For the proposed filter, the number of superpixel blocks was set to 3000 with a compactness parameter $\gamma = 40$, while a 3 × 3 × 3 neighborhood window was adopted and $T_{th} = -1$. For comparison, the V-BM3D method used a noise standard deviation $\sigma = 25$, and the 3D-CCM method employed a local 25 × 25 × 3 neighborhood window with $T_{th} = -1$.

The filtering results of different methods for the 212th

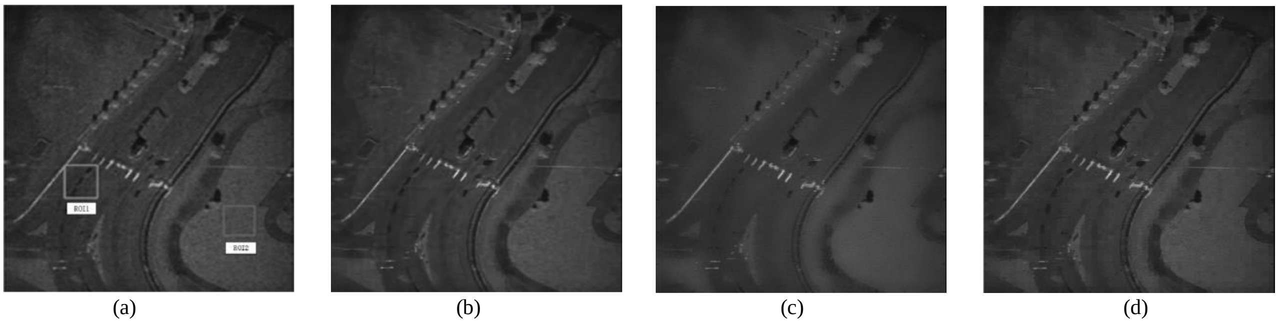

Fig. 5. Comparison of the effect of different filtering methods for the 212th frame of the test image. (a) Original image, (b) V-BM3D, (c) 3D-CCM, (d) Proposed method.

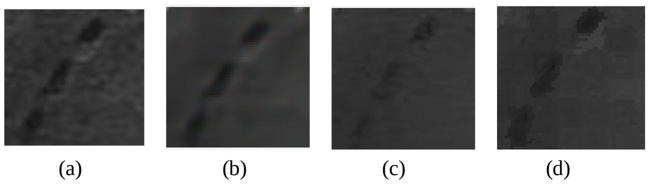

Fig. 6. ROI1 region, comparison of the effect of different filtering methods. (a) Original image, (b) V-BM3D, (c) 3D-CCM, (d) Proposed method.

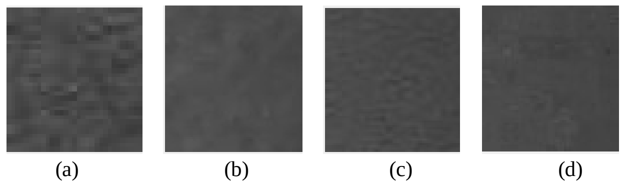

Fig. 7. ROI2 region, comparison of the effect of different filtering methods. (a) Original image, (b) V-BM3D, (c) 3D-CCM, (d) Proposed method.

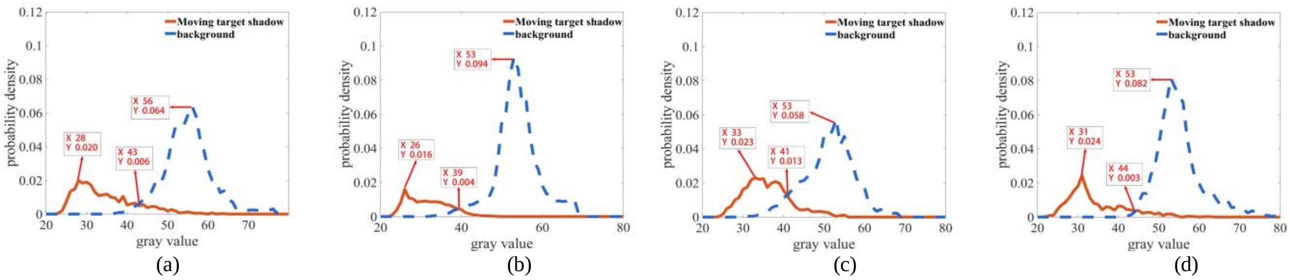

Fig. 8. Comparison of the gray-level probability density distributions of the shadow of moving target and the background in the ROI1 region. (a) Original image, (b) V-BM3D, (c) 3D-CCM, (d) Proposed method.

TABLE I. QUANTITATIVE EVALUATION INDICES FOR SPECKLE FILTERING

Filtering Method	Evaluation Metrics	
	ENL (ROI2)	FOM (ROI1)
Original	133	0.58
V-BM3D	532	0.55
3D-CCM	600	0.14
Proposed	487	0.64

frame are shown in Fig. 5. To evaluate the filtering performance and the retention effect of the shadow edge of the moving target, two regions of interest (ROIs) of size 50 × 50 pixels were selected, as shown in Fig. 6 and Fig. 7. ROI1 contains the moving target shadow, ROI2 is the background area. As shown in Table I, all methods effectively smooth the background speckle in ROI2, which is confirmed by the equivalent number of looks (ENL) [15] values. However, in ROI1, 3D-CCM suffers from severe motion blur and edge distortion, as indicated by a low figure of merit (FOM) [16], due to the misalignment of moving objects. In contrast, the proposed method achieves the highest FOM, which preserves sharp shadow edges and structural details through dynamic superpixel registration.

Fig. 8 presents a statistical comparison of the gray-level probability density distributions of the shadow of moving target and the background within ROI1. A wider separation between the shadow and background peaks indicates superior contrast and target detectability. Compared with V-BM3D and 3D-CCM, the proposed method achieves the greatest contrast disparity between the statistical distributions of the moving target shadow and the background. This result further validates that the proposed method effectively enhances shadow-to-background contrast, providing a robust statistical foundation for subsequent surveillance and detection tasks.

V. CONCLUSION

In this paper, a novel speckle noise filtering method for video SAR systems is proposed. The method employs the SLIC algorithm to segment SAR images into superpixels based on pixel intensity and spatial similarities within each frame, and spatially aligns the dynamic superpixels across frames. Subsequently, for each pixel, i.i.d. samples with similar scattering characteristics are identified within the aligned spatiotemporal superpixel domain and averaged to suppress speckle noise. Experimental results demonstrate that the proposed method effectively reduces speckle noise while preserving the sharp edges and structural details of moving target shadows. Furthermore, by enhancing the target-to-background contrast, the method improves the overall detection reliability for video SAR-GMTI.

ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China under Grant 62301398, and in part by the Hangzhou Youth Talent Promotion Project. (Corresponding author: Min Tian)

REFERENCES

[1] H. Wang, Z. Chen, and S. Zheng, "Preliminary research of low-RCS moving target detection based on Ka-band video SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 811–815, 2017.

[2] J. Chen, D. An, W. Wang, *et al.*, "A novel generation method of high-quality video image for high-resolution airborne ViSAR," *Remote Sens.*, vol. 13, no. 18, Art. no. 3706, 2021.

[3] A. Damini, B. Balaji, C. Parry, *et al.*, "A video SAR mode for the X-band wideband experimental airborne radar," in *Proc. SPIE, Algorithms for Synthetic Aperture Radar Imagery XVII*, vol. 7699, Orlando, FL, USA, 2010, Art. no. 76990E.

[4] J. Ding, "Focusing algorithms and moving target detection based on video SAR," *J. Radars*, vol. 9, no. 2, pp. 321–334, 2020.

[5] J. Li, "Research on coherent spot noise suppression of synthetic aperture radar images," Ph.D. dissertation, Natl. Univ. Defense Technol., Changsha, China, 2014.

[6] B. Zhou, C. Liu, and X. Zhou, "A video SAR moving target detection method based on shadow detection," *Microelectronics & Comput.*, vol. 38, no. 12, pp. 24–30, 2021.

[7] J. S. Lee, "Digital image smoothing and the sigma filter," *Comput. Vis. Graph. Image Process.*, vol. 24, no. 2, pp. 255–269, Nov. 1983.

[8] A. Buades, B. Coll, and J. M. Morel, "A non-local algorithm for image denoising," *in Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 60–65, 2005.

[9] J. Sun, Y. Bie, and H. Wang, "Based on video SAR target shadow detection method of the sequential images," *Aerosp. Shanghai*, vol. 38, no. S1, pp. 43–48, 2021.

[10] S. Liu, X. Cui, and S. Chen, "Speckle filtering of video SAR image based on space-time context information," *Syst. Eng. Electron.*, vol. 46, no. 2, pp. 446–458, 2024.

[11] R. Achanta *et al.*, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 5, pp. 2274–2282, Nov. 2012.

[12] J. Chen, Z. Li, and B. Huang, "Linear spectral clustering superpixel," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3317–3330, 2017.

[13] D. Xiang, T. Tang, L. Zhao, *et al.*, "Superpixel generating algorithm based on pixel intensity and location similarity for SAR image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1414–1418, 2013.

[14] F. Zhang, S. Lu, D. Xiang, and X. Yuan, "An improved superpixel-based CFAR method for high-resolution SAR image ship target detection," *J. Radars*, vol. 12, no. 1, pp. 120–139, 2023.

[15] F. Argenti, A. Lapini, T. Bianchi, and L. Alparone, "A tutorial on speckle reduction in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 3, pp. 6–35, Sep. 2013.

[16] S. W. Chen, "SAR image speckle filtering with context covariance matrix formulation and similarity test," *IEEE Trans. Image Process.*, vol. 29, pp. 6641–6654, 2020.